



AKTU

B.Tech : I-Year



Electronics Engg.

UNIT -2 : BJT and FET

Lecture - 1

Today's Target

- Bipolar Junction Transistor (BJT)
- DPP
- PYQs

By Gulshan sir



UNIT- 2

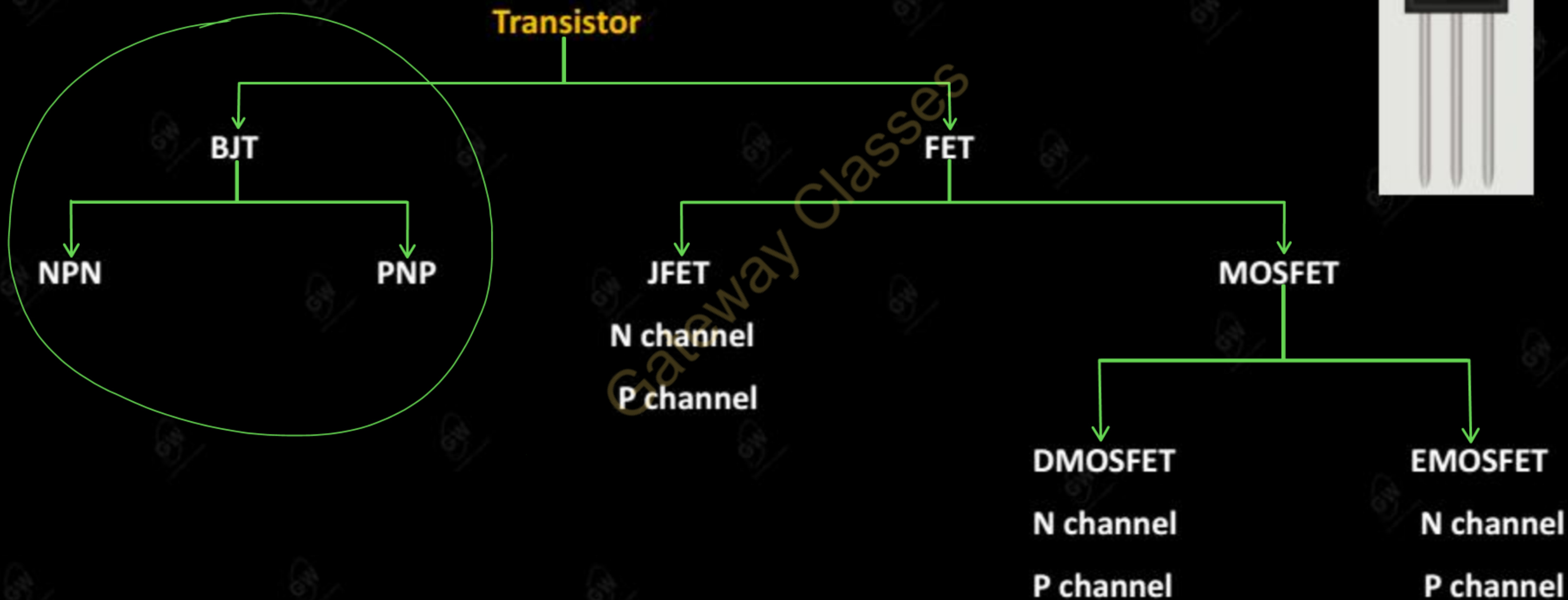
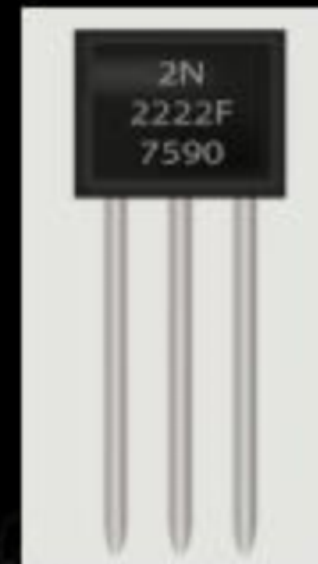
Bipolar Junction Transistor (BJT)

Transistor Construction, Operation, Amplification action. Common Base, Common Emitter, Common Collector Configuration

Field Effect Transistor (FET)

Construction and Characteristic of JFETs. Transfer Characteristic. MOSFET (MOS) (Depletion and Enhancement) Type, Transfer Characteristic.

- Transistor is a semiconductor device that can amplify or switch electronic signal and electrical power.
- Transistors are fundamental components in modern electronics.



BJT : Bipolar Junction Transistor

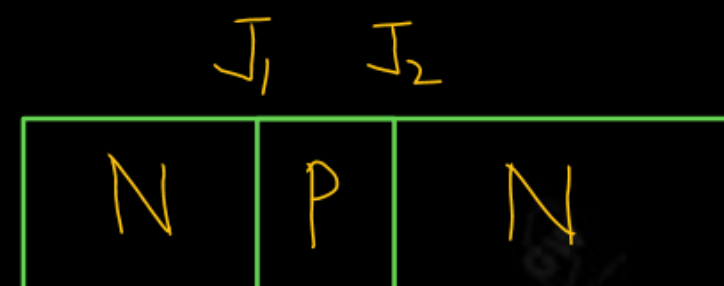
FET : Field Effect Transistor

JFET : Junction Field Effect Transistor

MOSFET : Metal oxide Semiconductor Field Effect Transistor

DMOSFET : Depletion MOSFET

EMOSFET : Enhancement MOSFET



Bipolar Junction Transistor (BJT) / Transistor

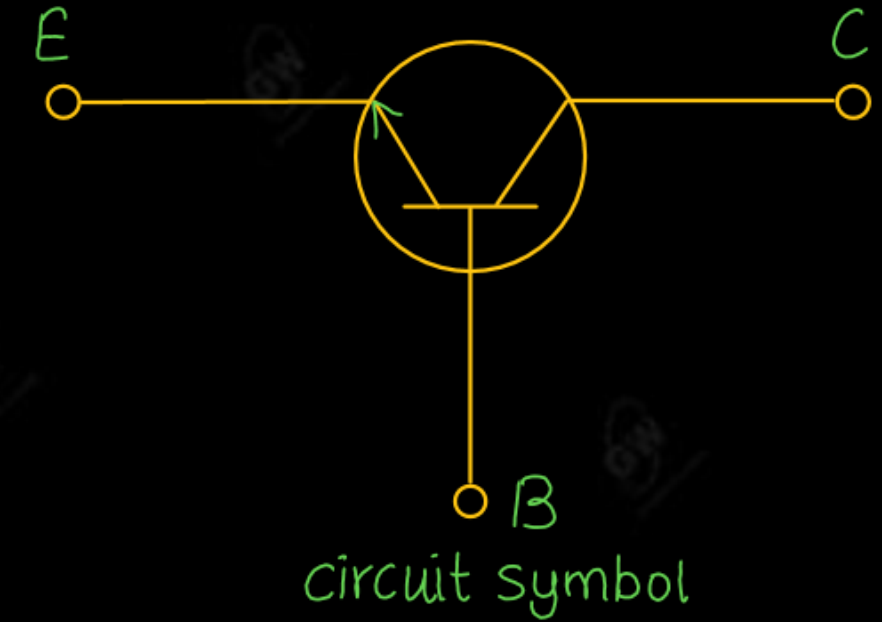
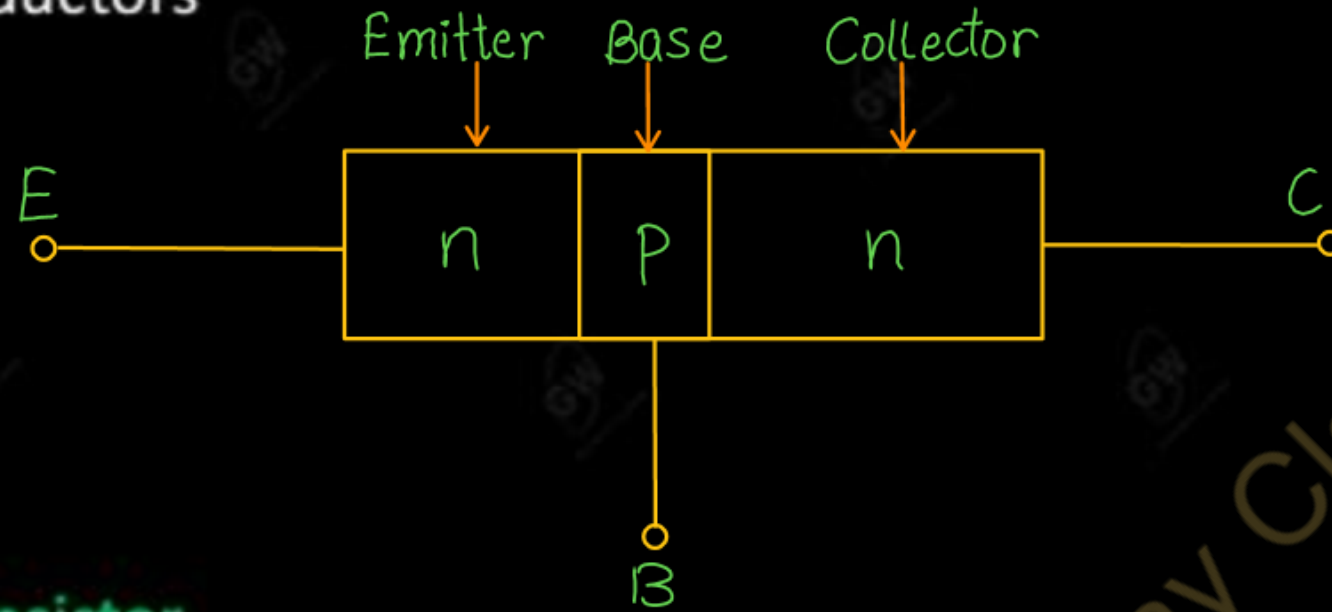
- A semiconductor device consists of two p-n junctions
- BJT consists of one n type material sandwiched between two p-type materials or one p-type material sandwiched between two n type materials.

Transistors are of two types

(i) NPN

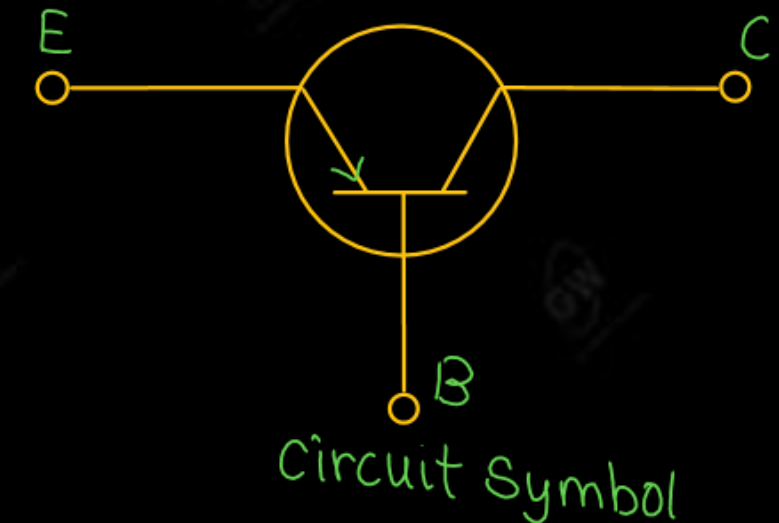
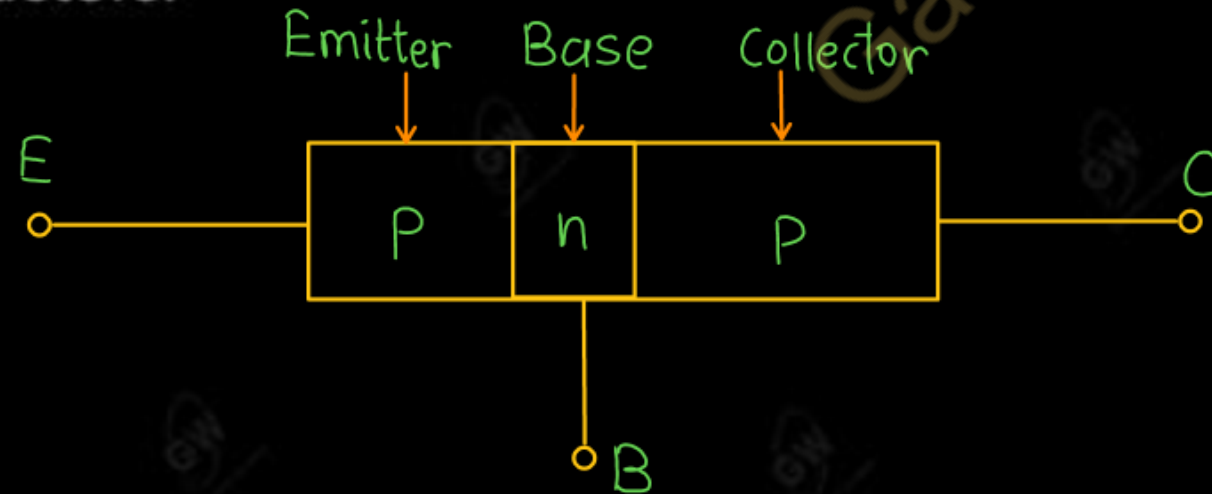
(ii) PNP

- It consists of a thin section of p-type semiconductor Sandwiched between two thicker sections of n-type semiconductors



PNP Transistor

- It consists of a thin section of n-type Semiconductor sandwiched between two thicker sections of p-type semiconductors.



Each type of transistor has three main parts

- (i) Emitter (E)
- (ii) Base (B)
- (iii) Collector (C)

Emitter

- It is a section on one side of the transistor
- It is of moderate size and heavily doped semiconductor
- It supply a large number of majority charge carriers (Electrons in npn and Holes in pnp transistors) for the flow of current through the transistor

Base

- It is the middle section
- It is very thin and lightly doped
- It controls the flow of majority charge carriers from emitter to collector

- It is a section on the other side of the transistor
- It is moderately doped and large in size as compared to the emitter.
- It collects the majority charge carriers for the circuit operation.

Area Profile of transistor

$$C > E > B$$

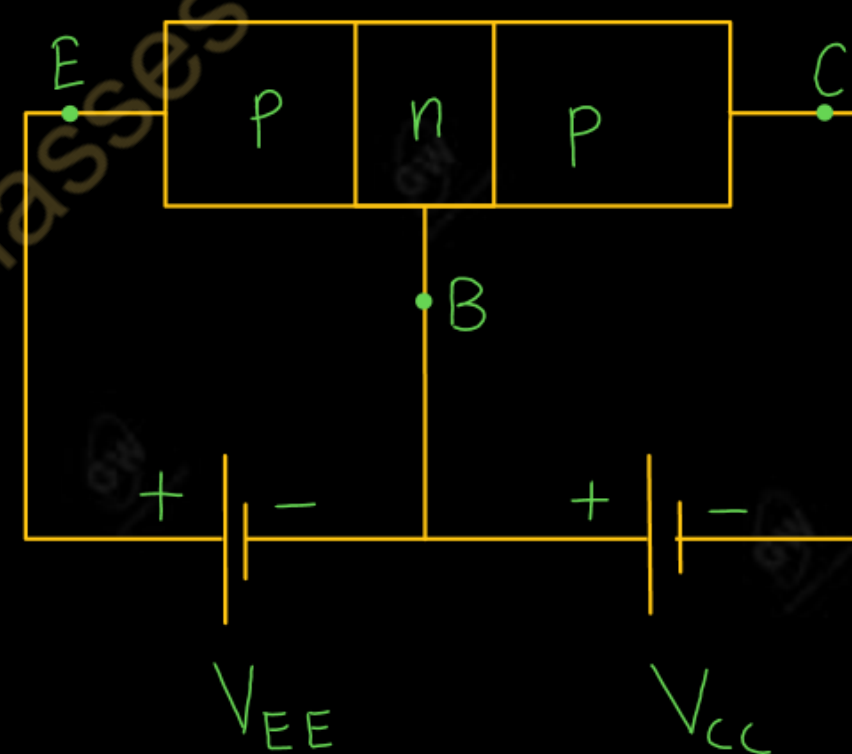
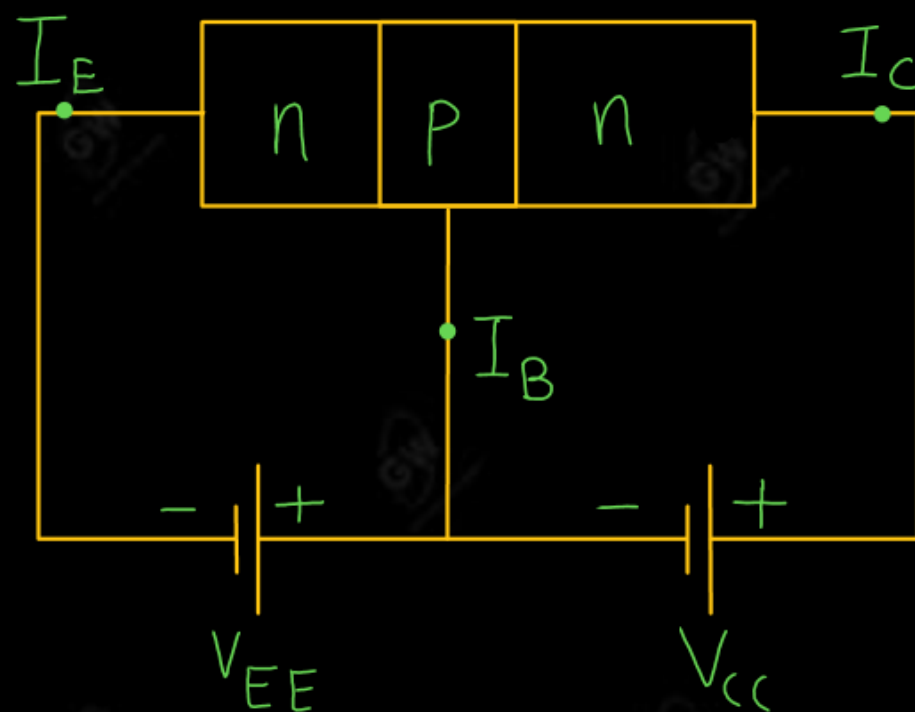
Doping Profile of transistor

$$E > C > B$$

Gateway Classes

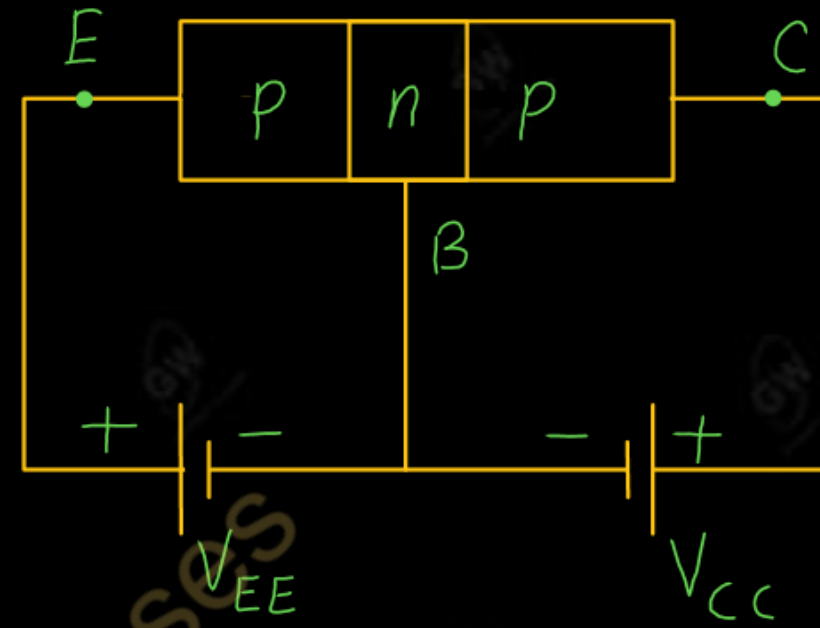
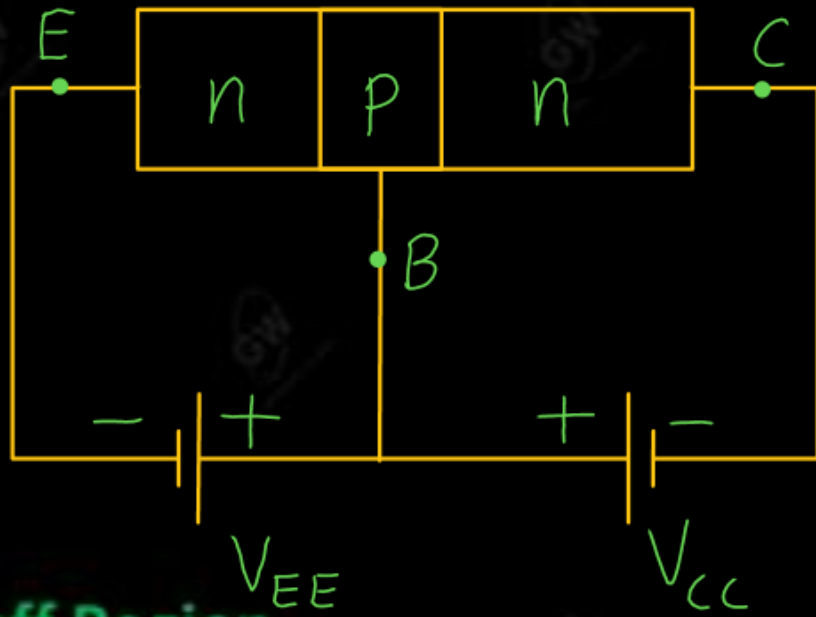
(i) Active Region

- In active region emitter-base junction is forward biased and collector-base junction is reverse biased.
- In this region transistor work as an amplifier.



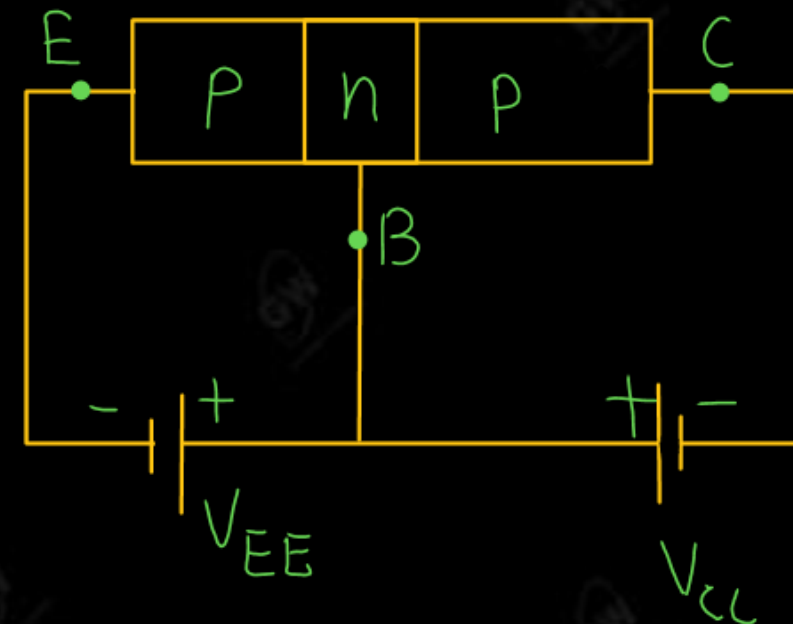
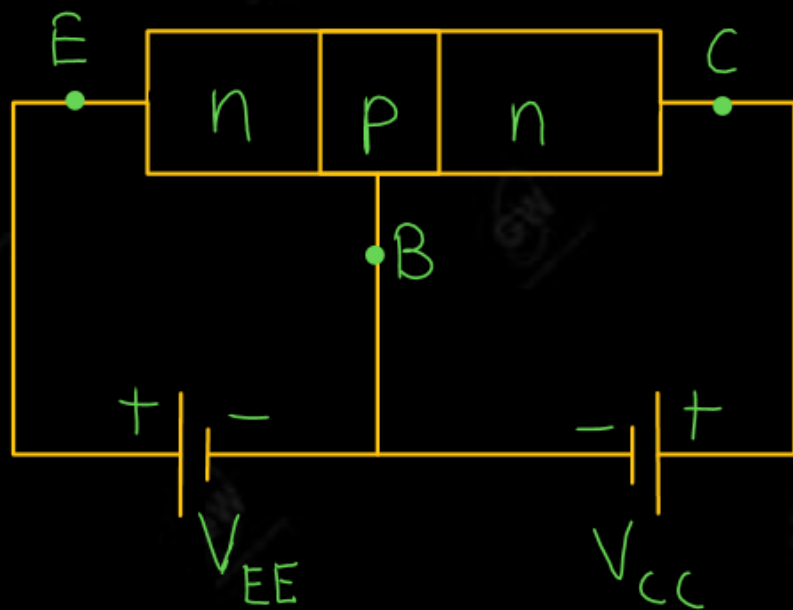
(ii) Saturation Region

- In this region emitter base junction and collector base junction are forward biased
- In this region transistor works as a closed switch



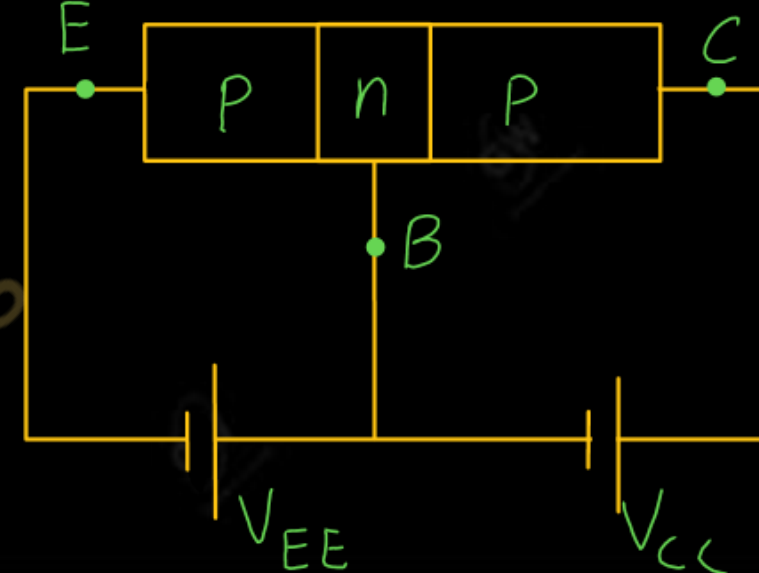
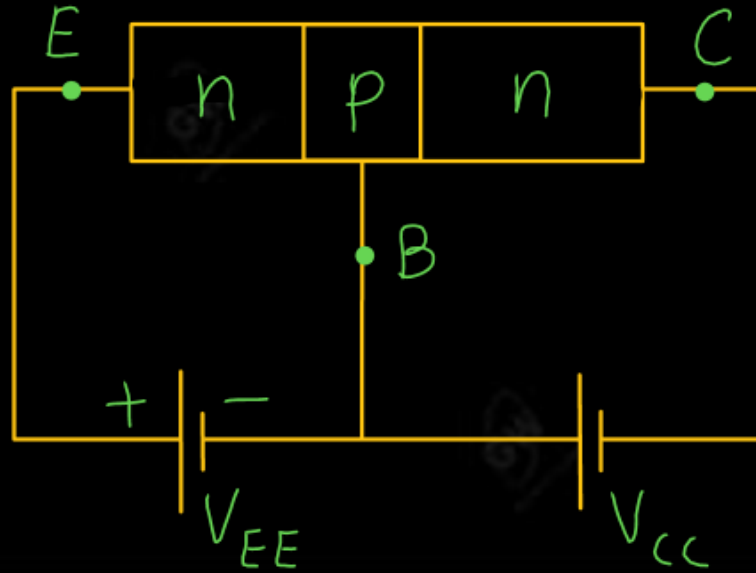
(iii) Cut-off Region

- In this region emitter base junction and collector base junction are reverse biased
- In this region transistor works as an open switch



(iv) Inverted Region

➤ In this region emitter base junction is reverse biased and collector base junction is forward biased



S.NO	Modes	Emitter bias junction	Collector bias junction	Application
1	Active	Forward	Reverse	Amplifier
2	Saturation	Forward	Forward	Closed switch (ON)
3	Cut off	Reverse	Reverse	Open switch (OFF)
4	Inverted	Reverse	Forward	Performance degradation

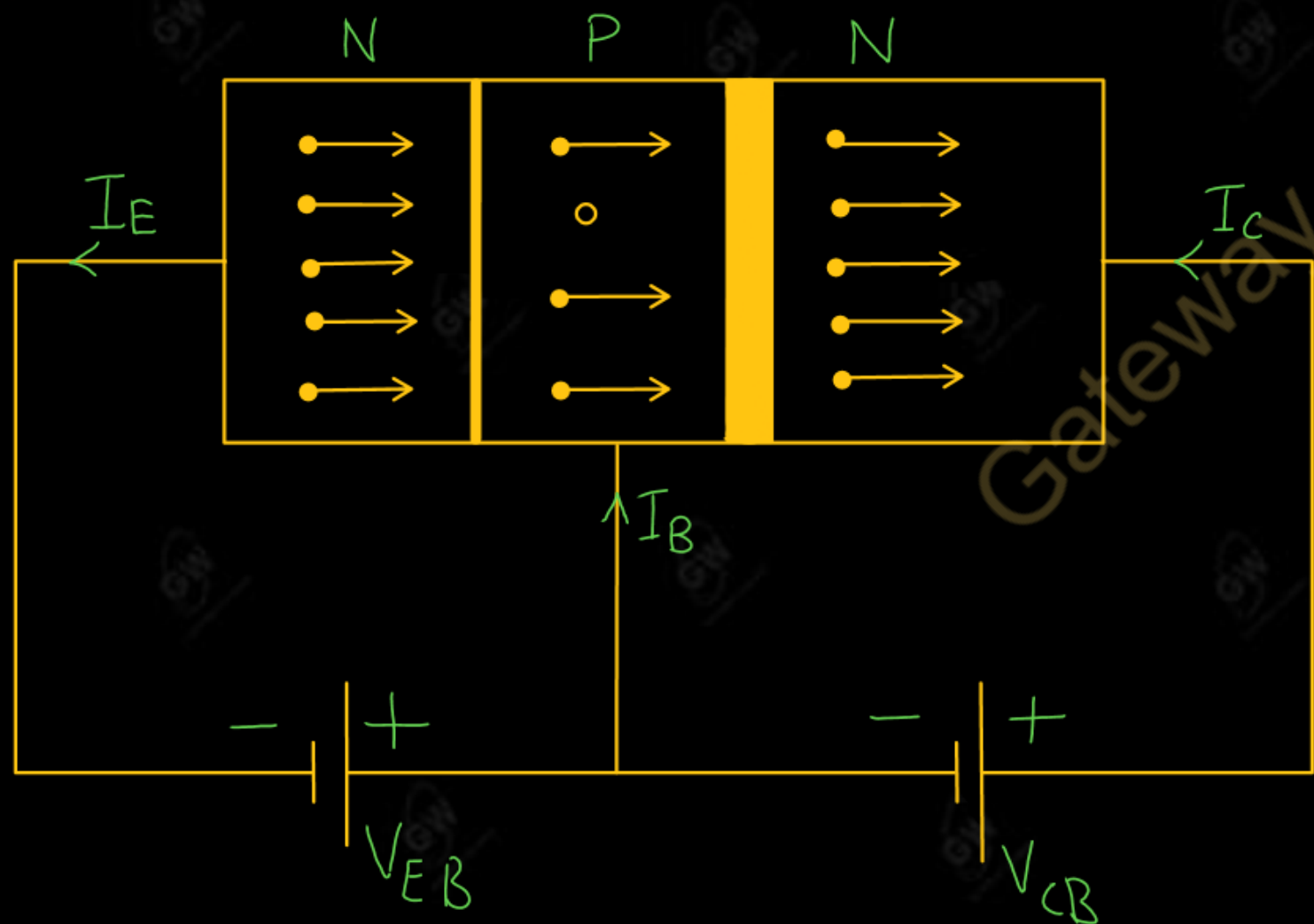
Operation of NPN Transistor in Active Region

OR

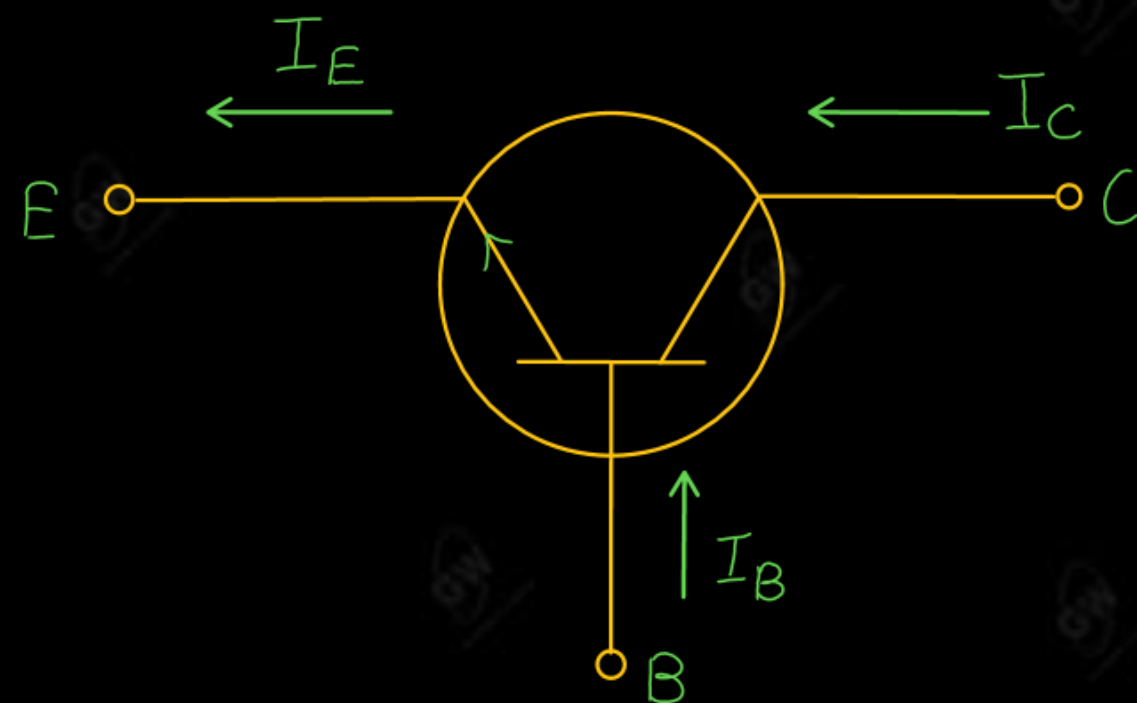
Working of NPN Transistor in Active Region

➤ To operate BJT in an active region

- (i) Emitter base junction is forward biased by battery V_{EB}
- (ii) Collector base junction is reverse biased by battery V_{CB}



$$I_E = I_B + I_C$$



- Width of depletion region for emitter base junction is small and for collector base junction is large
- Due to forward biased, a large number of electrons in the emitter are pushed towards the base which constitute the emitter current (I_E)
- These electrons pass through the base region and combine with holes.
- Since, base is lightly Doped and very thin, a few electrons ($< 5\%$) combine with the holes and constitute the base current (I_B)
- The remaining electrons ($> 95\%$) are pulled by the collector which is at high positive potential. The electrons are finally collected by positive terminal of battery V_{CB} and constitute collector current (I_C)
- Apply Kirchhoff's current law to the transistor

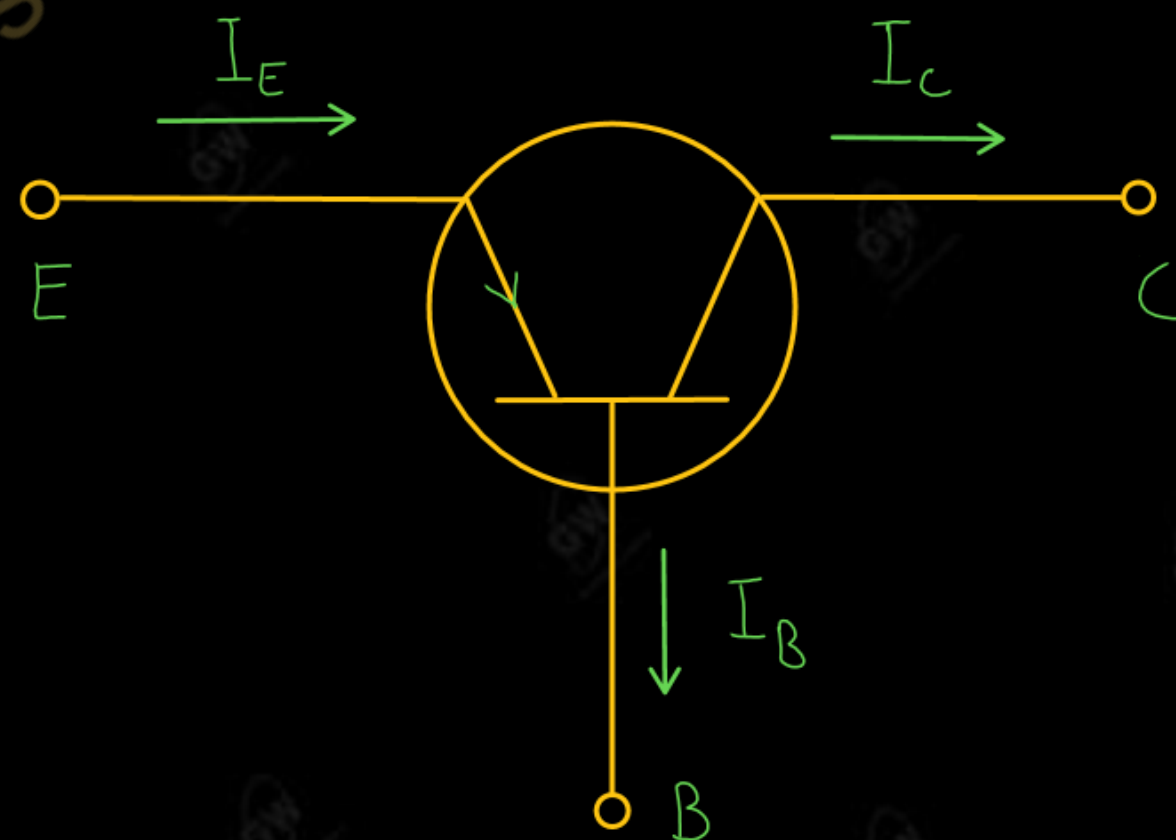
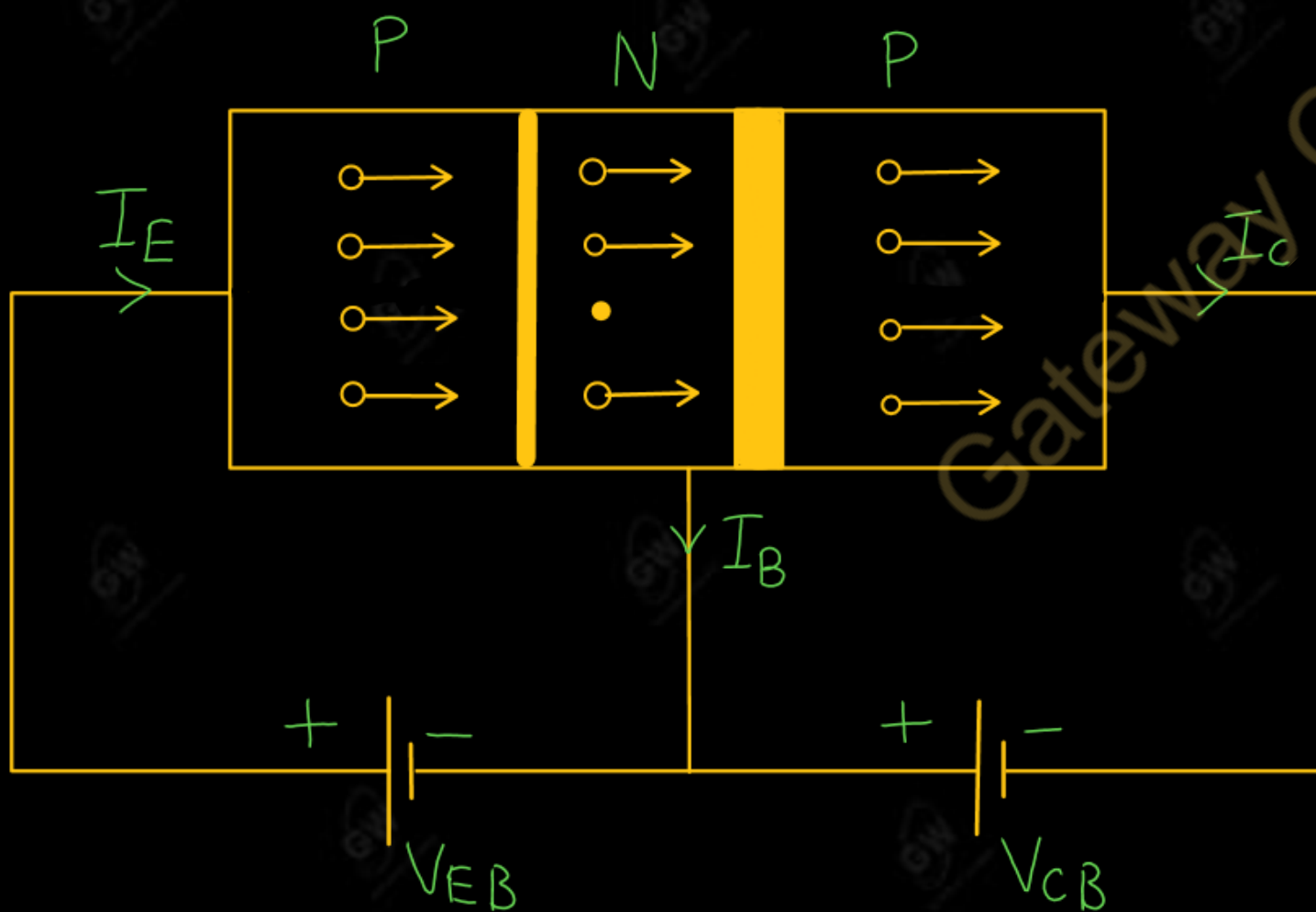
$$I_E = I_B + I_C$$

OR

Working of PNP Transistor in active region

➤ To operate BJT in an active region

- (i) Emitter base junction is forward biased by battery V_{EB}
- (ii) Collector base junction is reverse biased by battery V_{CB}



- Width of depletion region for emitter base junction is small and for collector base junction is large
- Due to forward biased, a large number of holes in the emitter are pushed towards the base which constitute the emitter current (I_E)
- These holes pass through the base region and combine with electrons.
- Since, base is lightly doped and very thin, a few holes ($< 5\%$) combine with the electrons and constitute the base current (I_B)
- The remaining holes ($> 95\%$) are pulled by the collector which constitute collector current (I_C)
- Apply Kirchhoff's current law to the transistor

$$I_E = I_B + I_C$$

BJT is a bipolar device

- BJT is a bipolar device because its operation depends on both majority and minority charge carriers.

BJT is a current controlled device

- BJT is a current controlled device because the current through two of its terminals is controlled by a current at the third terminal

Gateway Classes

➤ Transistor has three terminals

(i) Emitter (E)

(ii) Base (B)

(iii) Collector (C)

➤ But we require four terminals to connect the transistor in a circuit as an amplifier. Two for input and two for output.

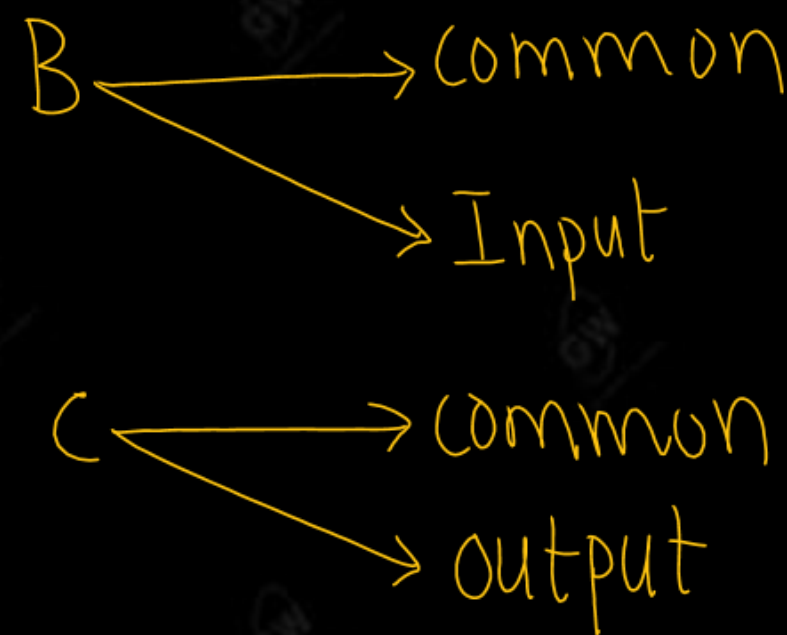
➤ This is achieved by making one terminal of transistor common to input and output.

➤ So, transistor has three configurations based on the common terminal.

(1) Common base configuration (CB)

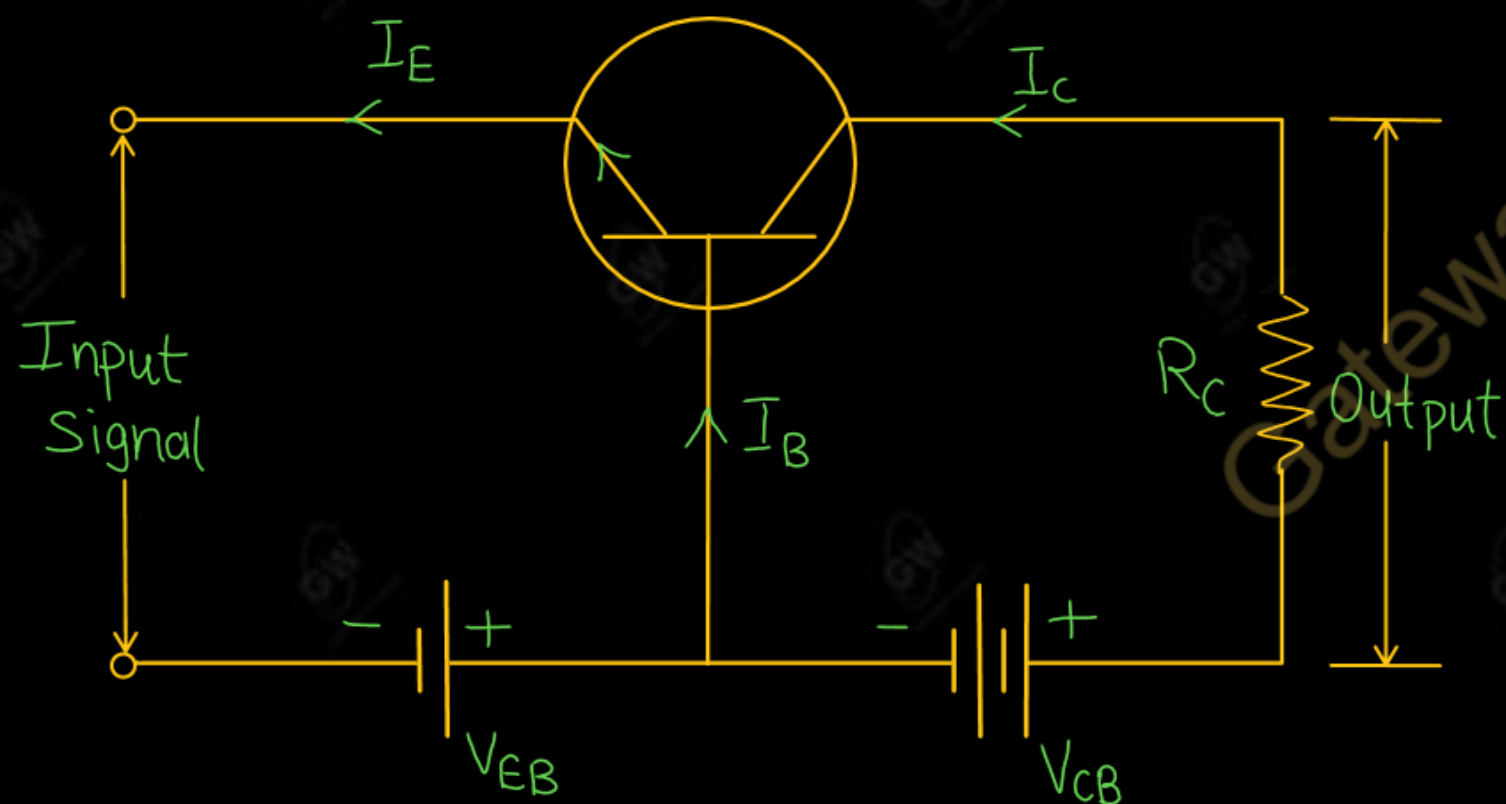
(2) Common emitter configuration (CE)

(3) Common collector configuration (CC)

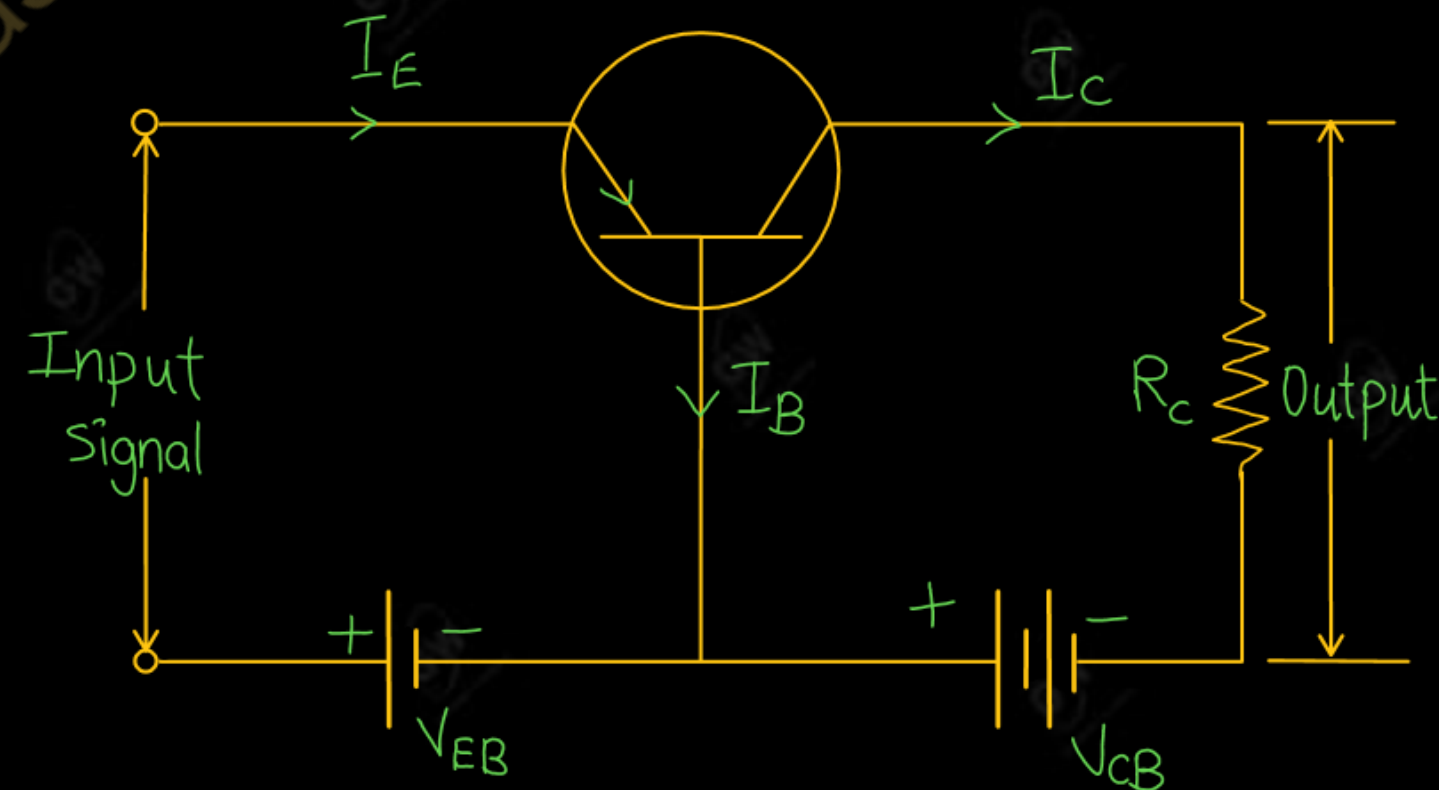


Common Base configuration (CB)

- Base of transistor is common to both input and output
- Input is applied between emitter and base
- Output is taken out from collector and base



Common Base NPN BJT



Common Base PNP BJT

(i) DC Current gain / Amplification factor (α)

- It is the ratio of output current (I_C) to the input current (I_E).

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E$$

- Since $I_E > I_C$, So value of α is less than 1 and no current gain is available in CB configuration.
- Value of α varies from 0.90 to 0.99

(ii) Expression for output current

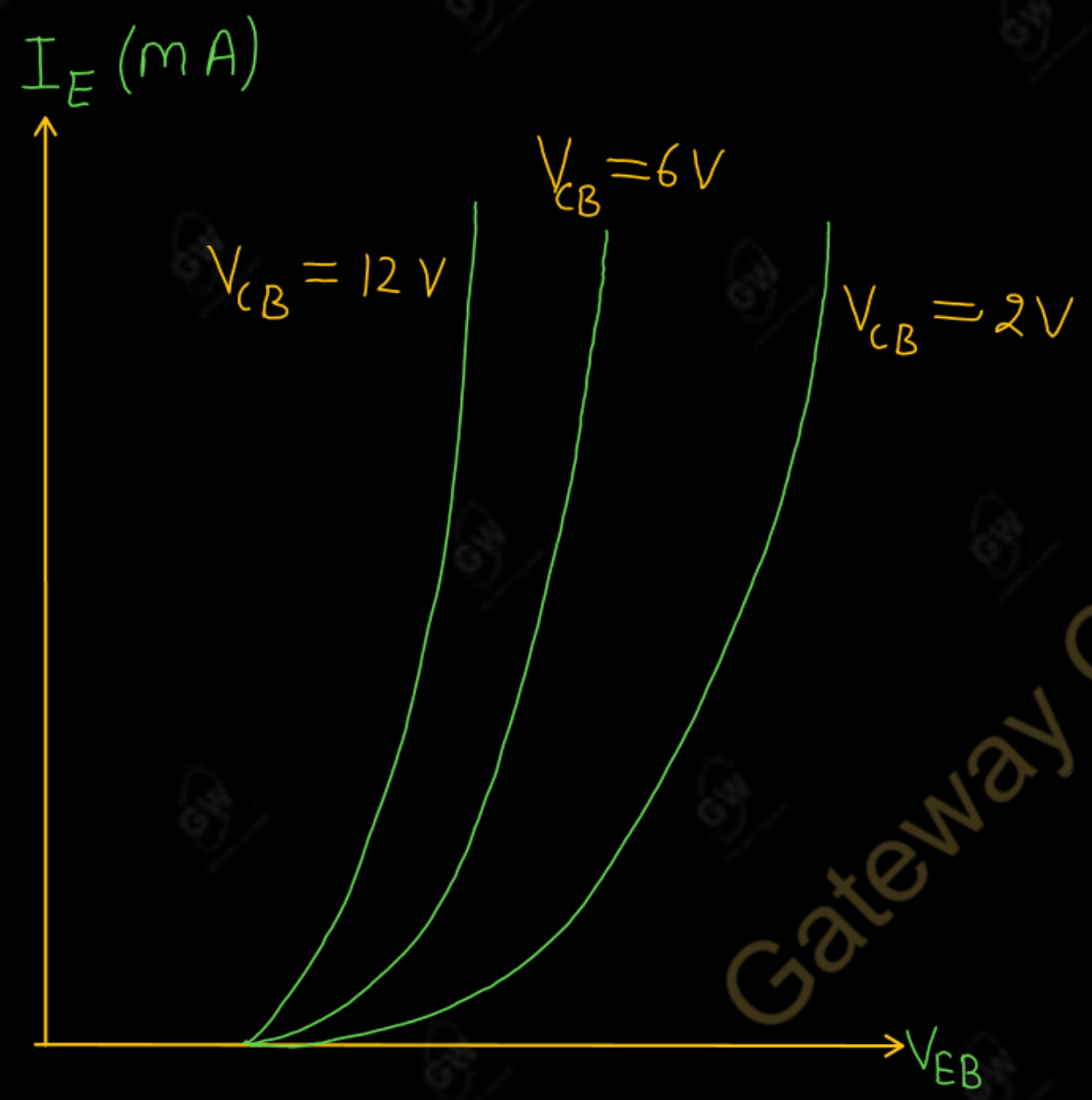
$$I_C = \alpha I_E + I_{CBO}$$

(iii) Input V-I Characteristics of CB Configuration

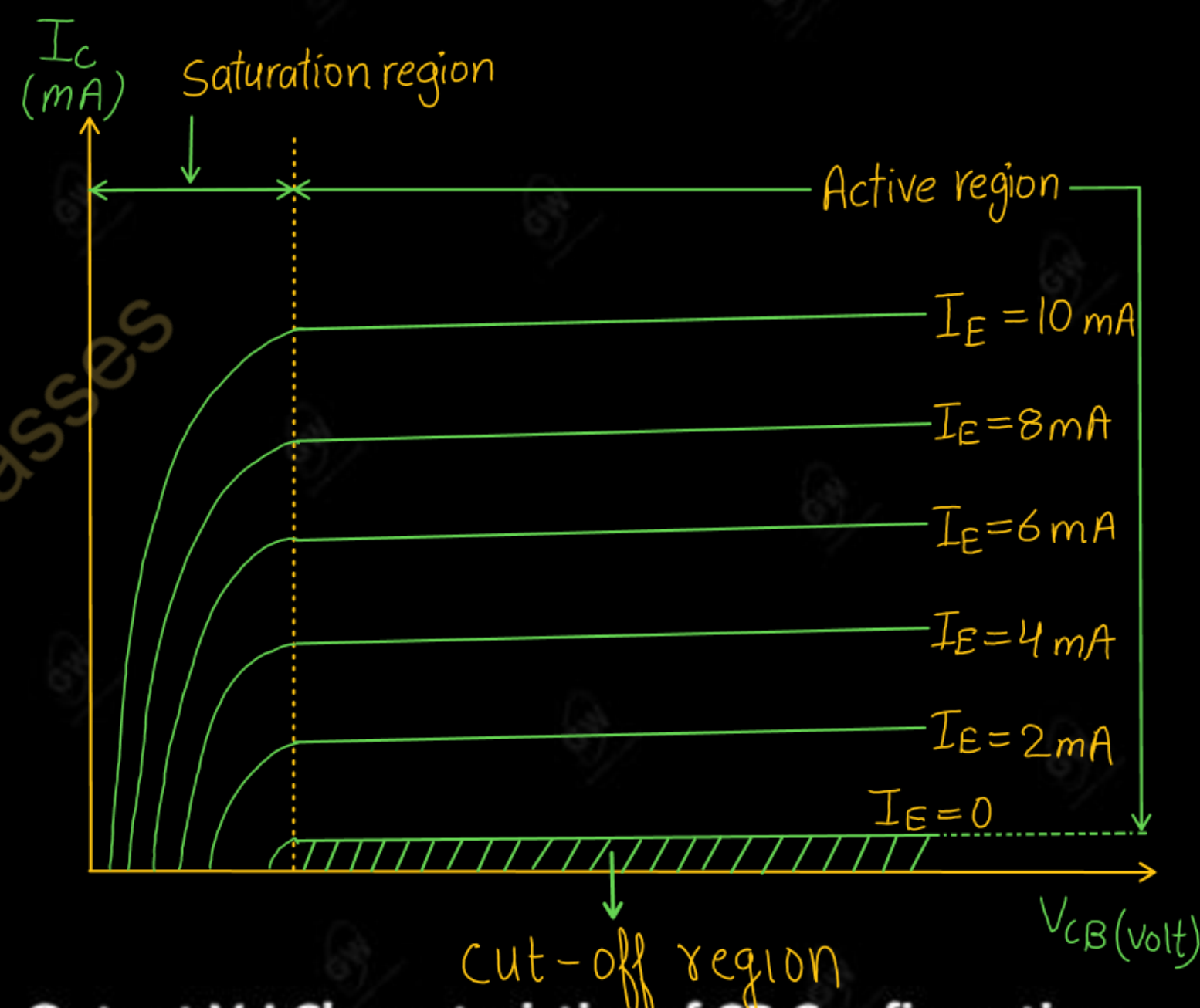
- It is graph between input current (I_E) and input voltage (V_{EB}) at constant output voltage (V_{CB})
- This graph is drawn for active region of BJT.

(iv) Output V-I Characteristics of CB Configuration

- It is graph between output current (I_C) and output voltage (V_{CB}) at constant input current (I_E).
- This graph is drawn for all three operating regions of BJT.



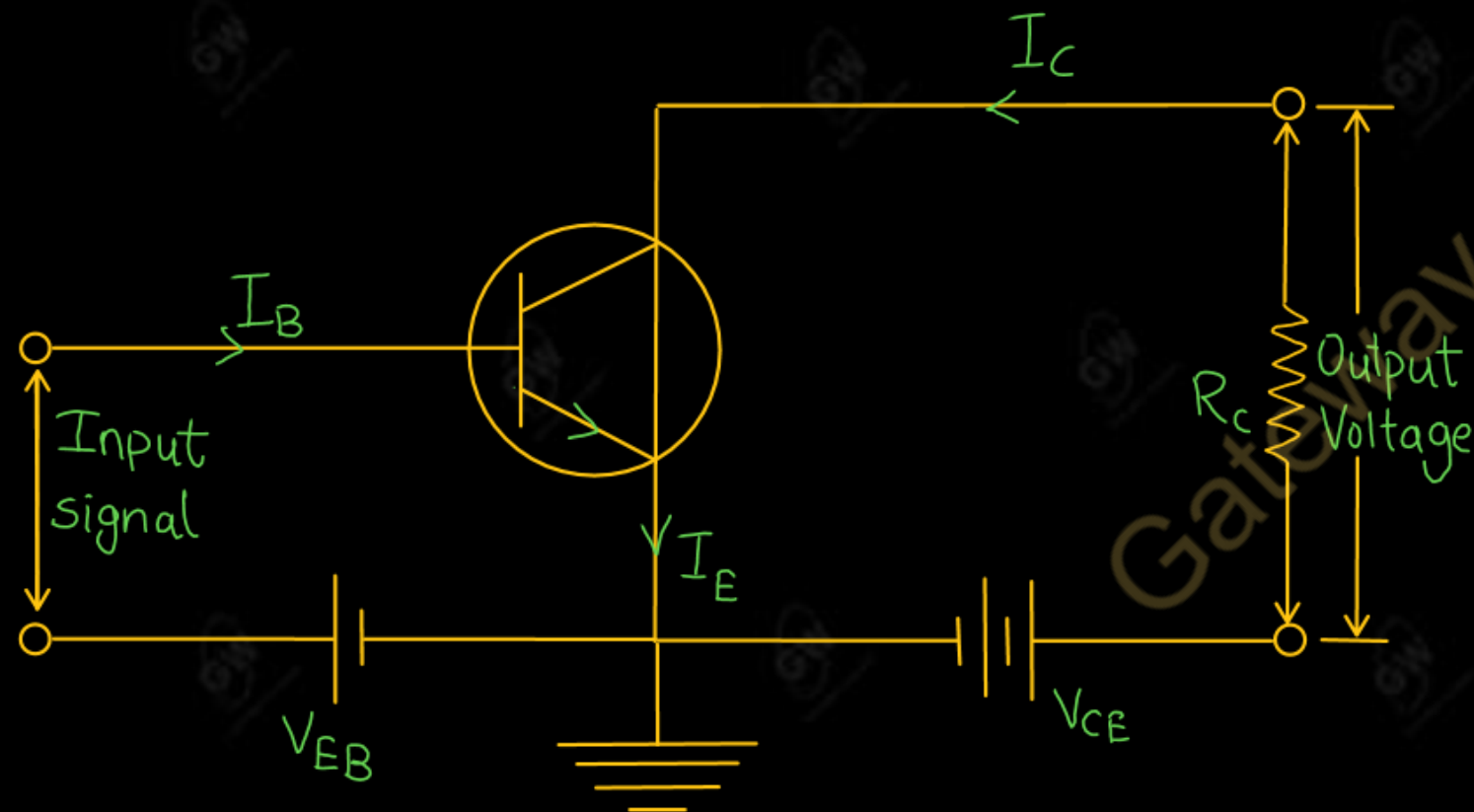
Input V-I Characteristics of CB Configuration



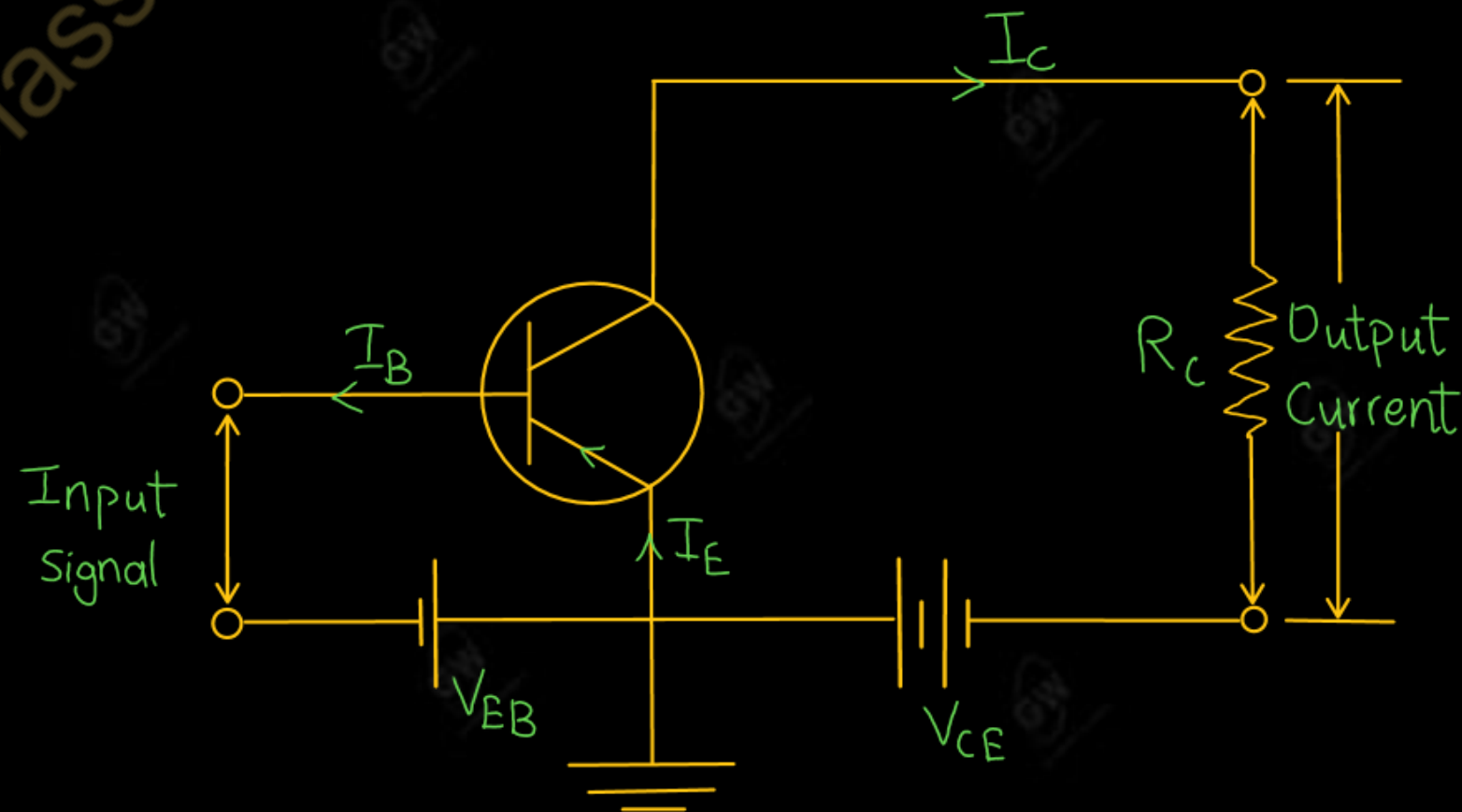
Output V-I Characteristics of CB Configuration

Common Emitter configuration

- Emitter of transistor is common to both input and output.
- Input is applied between base and emitter
- Output is taken out from collector and emitter



Common Emitter NPN BJT



Common Emitter PNP BJT

(i) DC Current gain / Amplification factor (β)

- It is the ratio of output current (I_C) to the input current (I_B).

$$\beta = \frac{I_C}{I_B}$$

- Since value of $I_B \ll I_C$, So $\beta \gg 1$ and current gain is available in CE configuration.
- Value of β varies from 20 to 500.

(ii) Expression for output current

$$I_C = \beta I_B + I_{CEO}$$

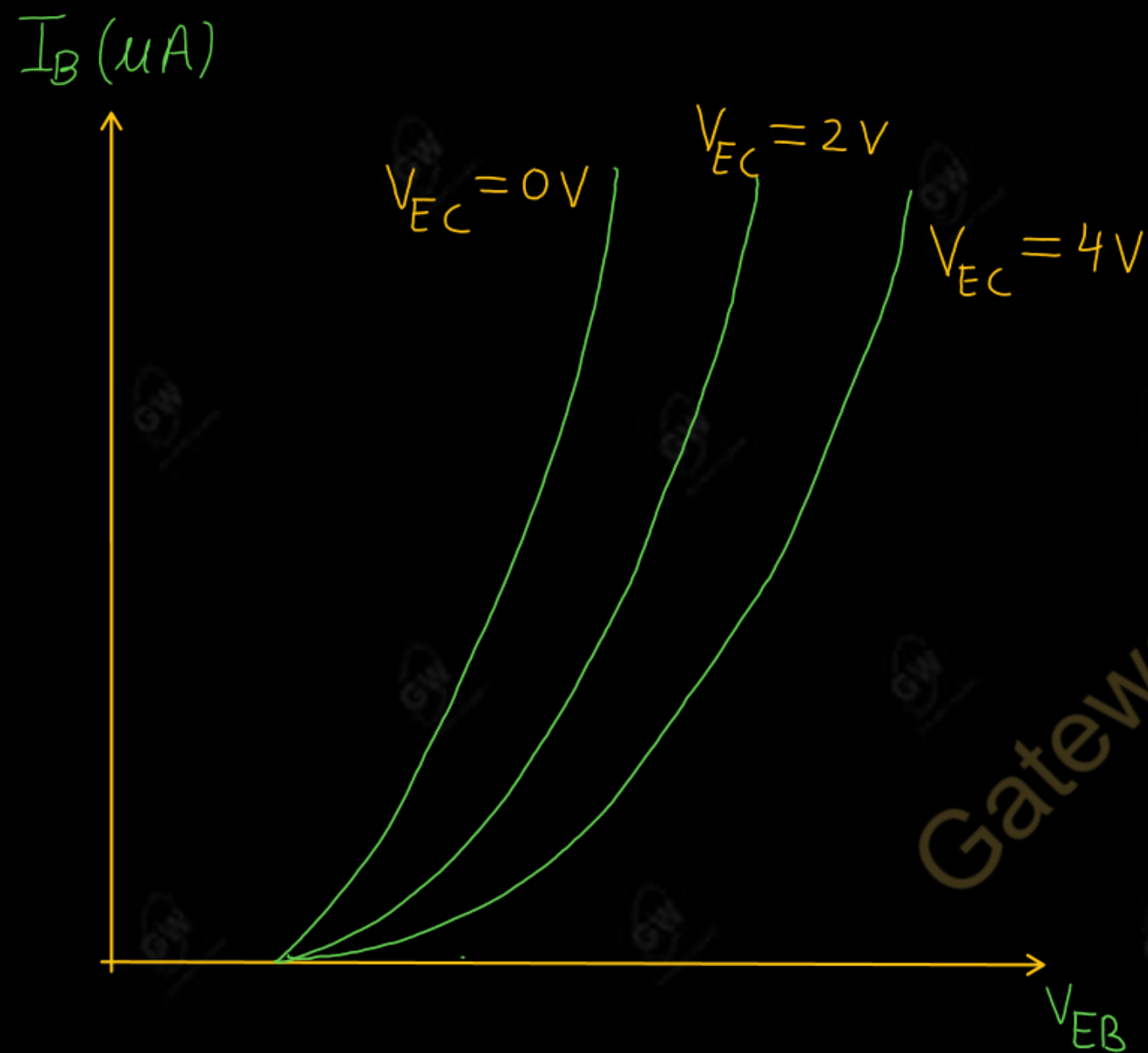
$$\{\text{Here } I_{CEO} = (\beta + 1) I_{CBO}\}$$

(iii) Input V-I Characteristics of CE Configuration

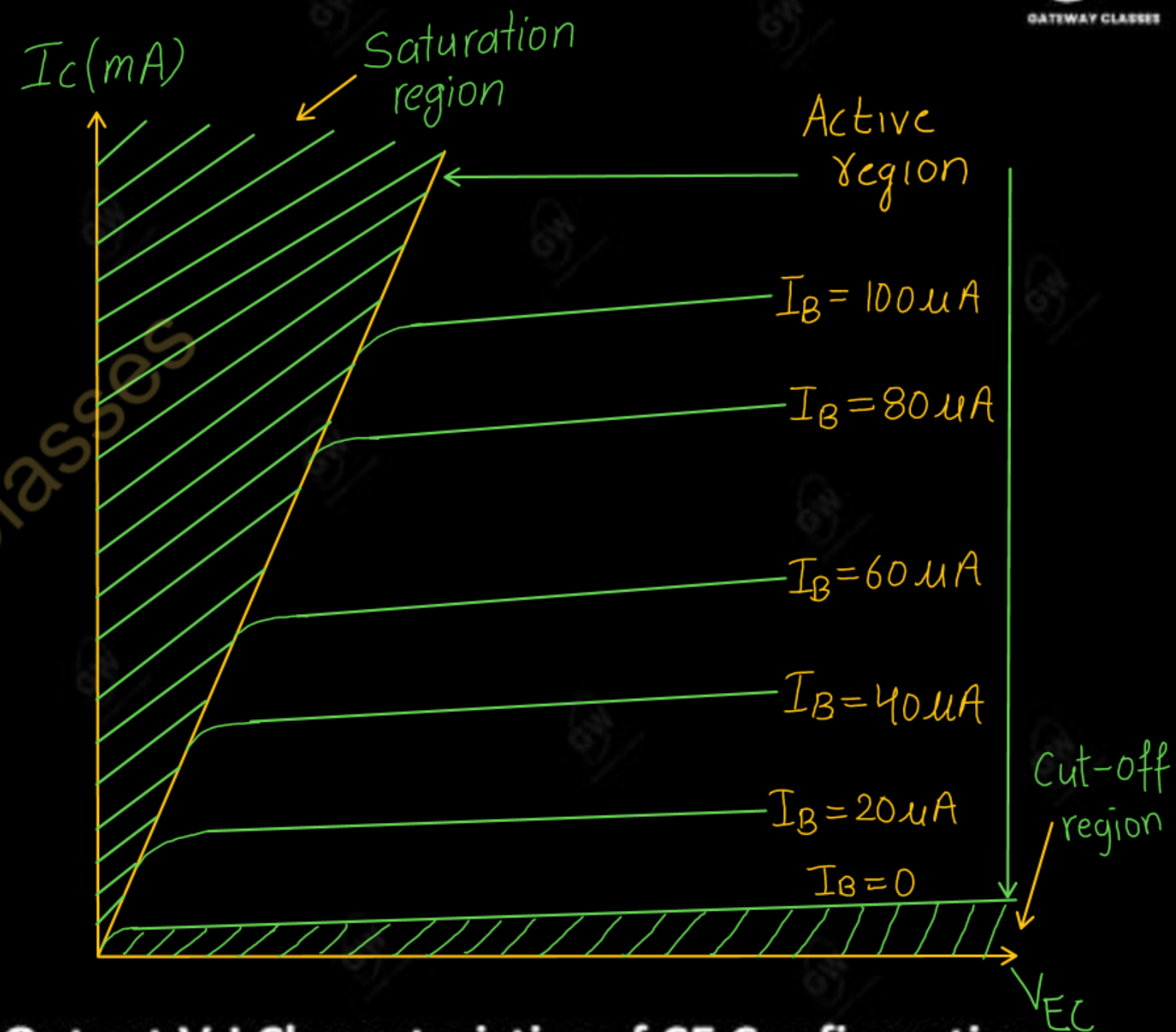
- It is graph between input current (I_B) and input voltage (V_{BE}) at constant output voltage (V_{CE}).
- This graph is drawn for active region of BJT.

(iv) Output V-I Characteristics of CE Configuration

- It is graph between output current (I_C) and output voltage (V_{CE}) at constant input current (I_B).
- This graph is drawn for all three operating regions of BJT



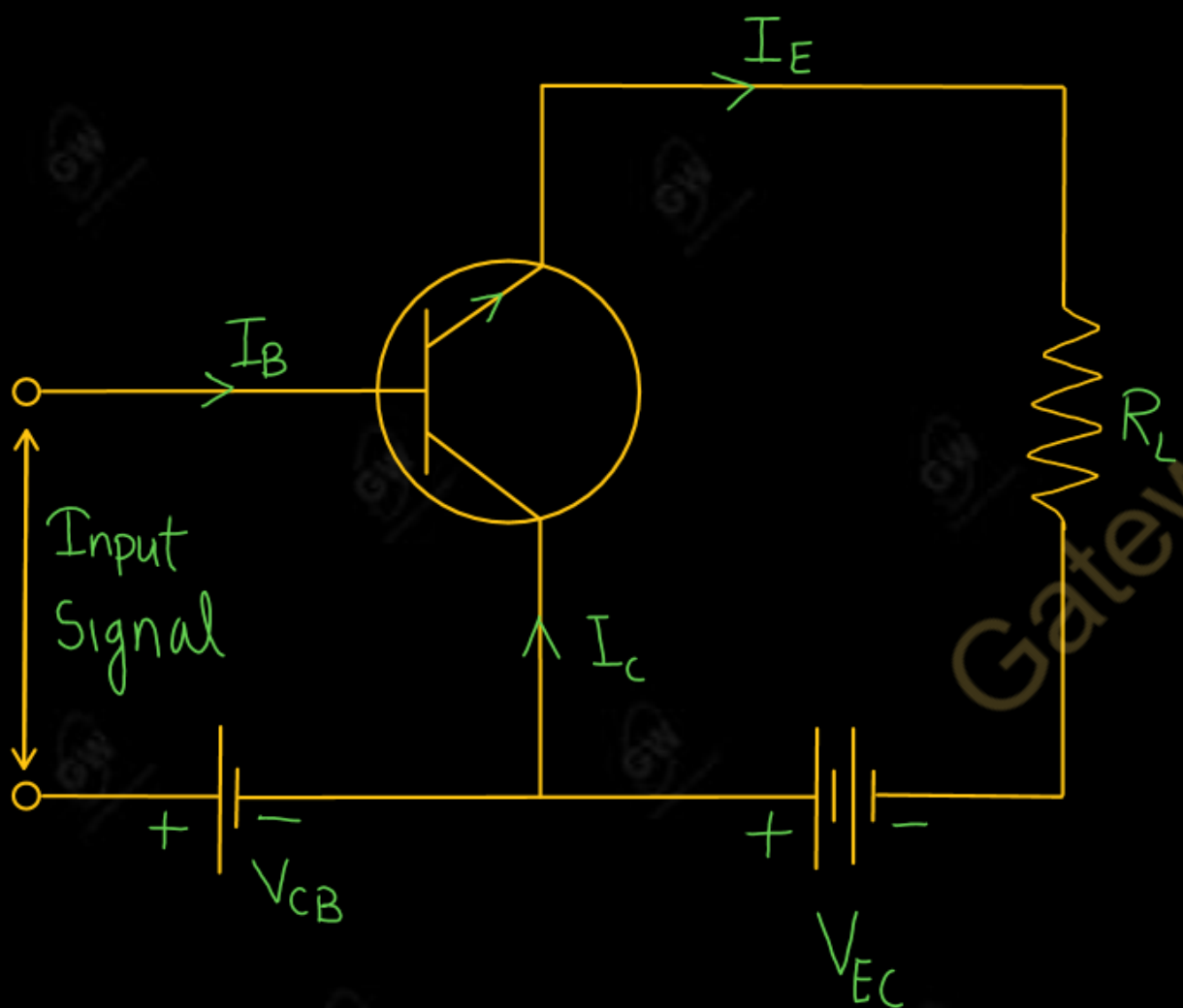
Input V-I Characteristics of CE Configuration



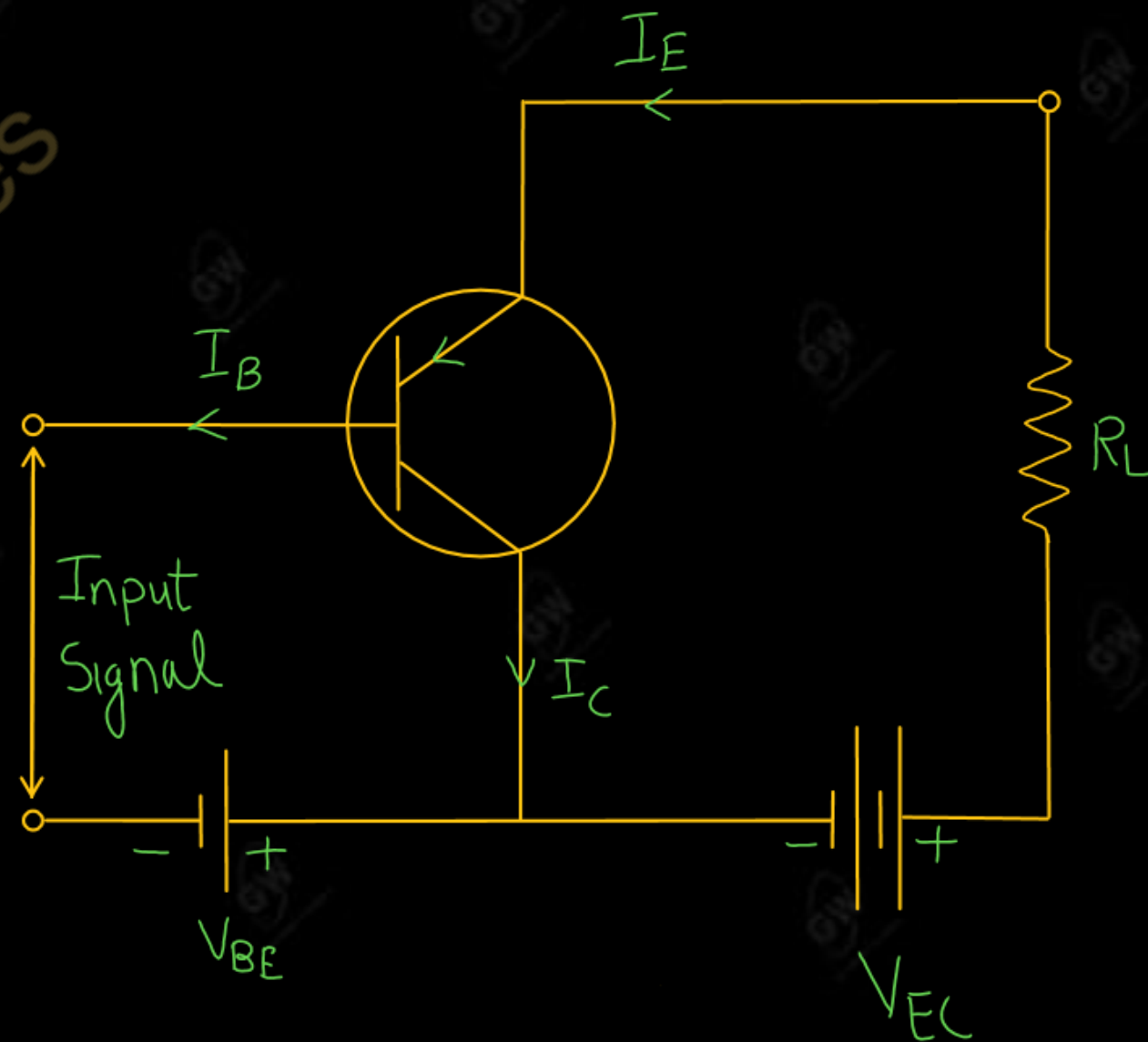
Output V-I Characteristics of CE Configuration

Common collector configuration (CC)

- Collector of transistor is common to both input and output
- Input is applied between base and collector
- Output is taken out from collector and emitter



Common Collector NPN BJT



Common Collector PNP BJT

(i) DC Current gain / Amplification factor (γ)

- It is the ratio of output current (I_E) to the input current (I_B).

$$\gamma = \frac{I_E}{I_B}$$

- Since $I_B \ll I_E$, So $\gamma \gg 1$ and current gain is available in CC configuration
- But voltage gain is not available in CC

(ii) Expression for output current

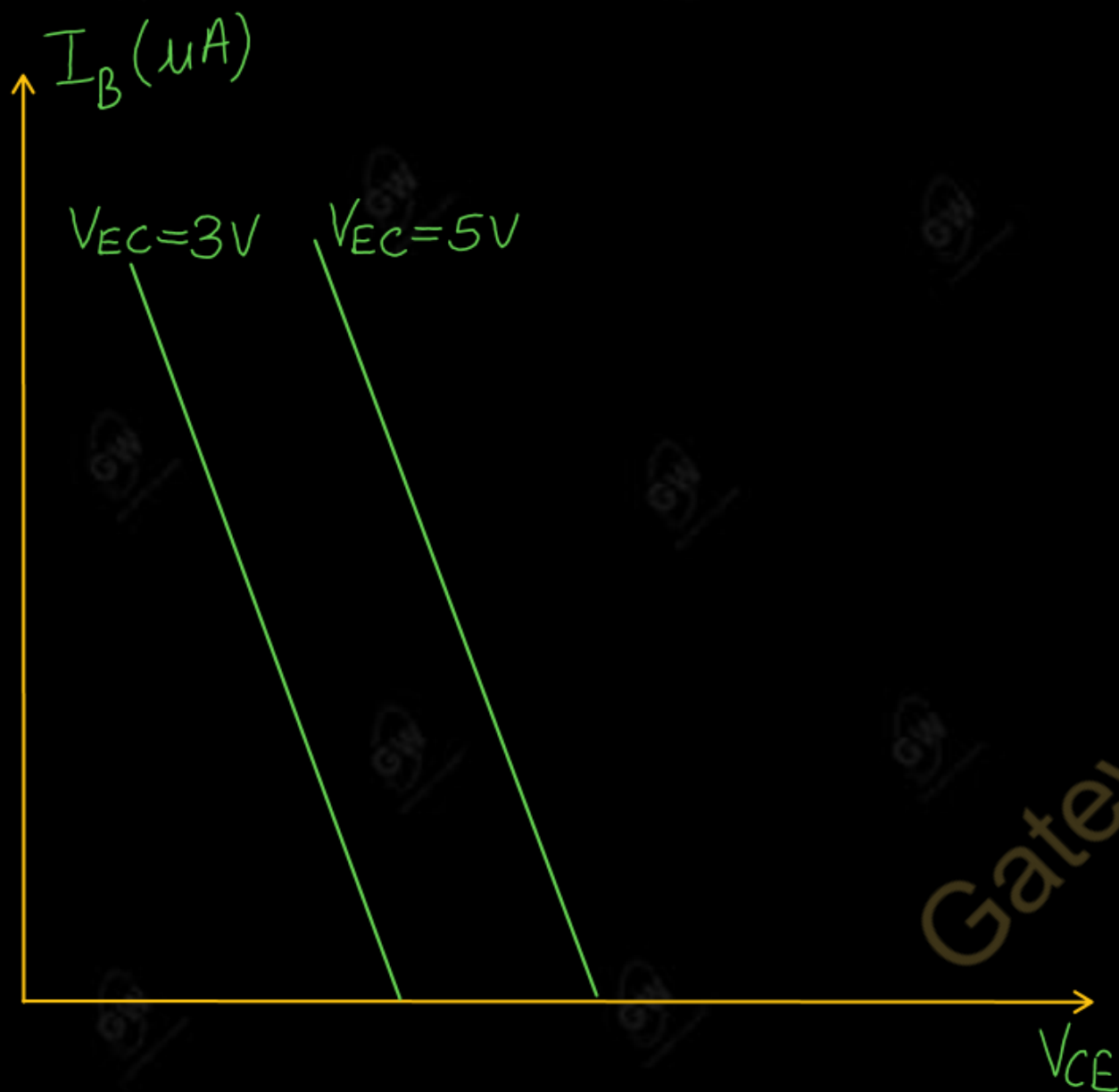
$$I_E = \gamma I_B + \gamma I_{CBO}$$

(iii) Input V-I Characteristics of CC Configuration

- It is graph between input current (I_B) and input voltage (V_{BE}) at constant output voltage (V_{CE})
- This graph is drawn for active region of BJT.

(iv) Output V-I Characteristics of CC Configuration

- It is graph between output current (I_E) and output voltage (V_{CE}) at constant input current (I_B).
- This graph is drawn for all three operating regions of BJT



Input V-I Characteristics of CC Configuration



Output V-I Characteristics of CC Configuration

We know that

$$I_E = I_B + I_C$$

Divide above equation by I_C

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\alpha} = \frac{1 + \beta}{\beta}$$

$$\alpha = \frac{\beta}{\beta + 1} \quad \checkmark \checkmark$$

$$\alpha(\beta + 1) = \beta$$

$$\alpha\beta + \alpha = \beta$$

$$\beta - \alpha\beta = \alpha$$

$$\beta(1 - \alpha) = \alpha$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{I_C}{I_E}$$

$$\beta = \frac{I_C}{I_B}$$

We know that

$$I_E = I_B + I_C$$

Divide above equation by I_B

$$\frac{I_E}{I_B} = \frac{I_B}{I_B} + \frac{I_C}{I_B}$$

$$\gamma = 1 + \beta$$

$$\gamma = \beta + 1$$

$$\beta = \gamma - 1$$

$$\beta = \frac{I_C}{I_B}$$

$$\gamma = \frac{I_E}{I_B}$$

Q.1 For a transistor $I_E = 10 \text{ mA}$ and $\alpha = 0.987$. Find I_C and I_B

Given

$$I_E = 10 \text{ mA}$$

$$\alpha = 0.987$$

We know that

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E$$

$$I_C = 0.987 \times 10$$

$$I_C = 9.87 \text{ mA}$$

$$I_E = I_C + I_B$$

$$\begin{aligned} I_B &= I_E - I_C \\ &= 10 - 9.87 \end{aligned}$$

$$I_B = 0.13 \text{ mA}$$

Q.2 Determine β , if $I_E = 5 \text{ mA}$, $I_C = 4.95 \text{ mA}$

Given

$$I_E = 5 \text{ mA}$$

$$I_C = 4.95 \text{ mA}$$

$$\alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{4.95}{5}$$

$$\alpha = 0.99$$

We know that

$$\beta = \frac{\alpha}{1-\alpha}$$

$$= \frac{0.99}{1-0.99}$$

$$\beta = 99$$

Q.3 In NPN transistor with $\beta = 98$ is operated in the CB configuration. If the emitter current is 2 mA and reverse saturation current is $12 \mu\text{A}$. What are the base and collector current?

Given

$$\beta = 98$$

$$I_E = 2 \text{ mA}$$

$$I_{CBO} = 12 \mu\text{A}$$

We know that

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\alpha = \frac{98}{98 + 1}$$

$$\alpha = 0.989$$

For CB

$$I_C = \alpha I_E + I_{CBO}$$

$$I_C = 0.989 \times 2 \times 10^{-3} + 12 \times 10^{-6}$$

$$I_C = 10^{-3} (0.989 \times 2 + 12 \times 10^{-3})$$

$$I_C = 1.99 \text{ mA}$$

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$$I_E = I_B + I_C$$

$$I_B = I_E - I_C$$

$$I_B = 2 - 1.99$$

$$I_B = 0.01 \text{ mA}$$

Q.4 In CB configuration, the value of $\alpha = 0.98$. A voltage drop of 4.9V is obtained across the resistor of $5K\Omega$ when connected in collector circuit. Find the base current

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Given

$$\alpha = 0.98$$

$$V = 4.9V$$

$$R = 5 \times 10^3 \Omega$$

By Ohm's law

$$I_c = \frac{V}{R}$$

$$I_c = \frac{4.9}{5 \times 10^3}$$

$$I_c = 0.98 \text{ mA}$$

$$\alpha = \frac{I_c}{I_E}$$

$$I_E = \frac{I_c}{\alpha}$$

$$I_E = \frac{0.98}{0.98}$$

$$I_E = 1 \text{ mA}$$

$$I_E = I_B + I_c$$

$$I_B = I_E - I_c$$

$$I_B = 1 - 0.98$$

$$I_B = 0.02 \text{ mA}$$

UNIT - 2 : DPP - 1

- Q.1** Explain why BJT is a Bipolar device & current controlled device ? **AKTU 2024**
- Q.2** Describe the construction of npn BJT. **AKTU 2021**
- Q.3** Explain the operation of NPN transistor in active region. **AKTU 2021**
- Q.4** Explain the working of N-P-N Common Base circuit with its circuit diagram. Draw the input and output characteristics of common base (CB) configuration. **AKTU 2023**
- Q.5** Explain the working of P-N-P Common Base circuit with its circuit diagram. Draw the input and output characteristics of common base (CB) configuration. **AKTU 2024**
- Q.6** Describe the construction and working of a NPN transistor in CE configuration w.r.t to size and doping. Also draw the input and output characteristics graph. **AKTU 2022**
- Q.7** Discuss the common collector (CC) configuration of a BJT. **AKTU 2024**
- Q.8** Define α and β with respect to BJT and derive the relationship between them **AKTU 2022**
- Q.9** A transistor having $\alpha = 0.975$ and $I_{CBO} = 10 \mu A$ is operated in CE mode. What is for the configuration? If the I_B is $250 \mu A$. Then calculate I_E and I_C **AKTU 2022**



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